

FINAL EXAMINATION

Course: **OBJECT ORIENTED PROGRAMMING**

Time: **100 minutes**

Term: **3 – Academic year: 2019-2020**

Lecturer(s): **Bui Tien Len, Nguyen Van Vu, Dinh Ba Tien, Nguyen Minh Huy, Trương Toàn Thịnh, Trần Duy Quang**

Student name:

Student ID:

(Notes: Closed book exam)

Question 1. a) What are class and object? Please give one example for each concept.

b) List and explain the usage of three access specifiers in C++?

c) Fill in the blanks:

(1)_____ lets us create a new class by reusing (2)_____ and (3)_____ from another class.

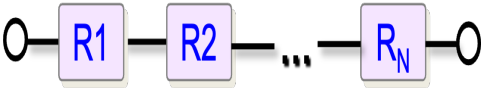
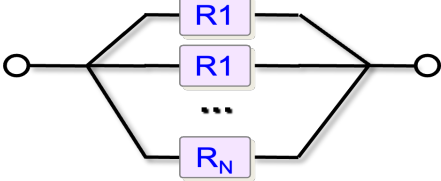
d) Give one purpose of Encapsulation (Data Hiding).

Question 2. Assume that all necessary libraries are included, read the C++ code below and answer the following questions:

<pre> 01 class Person { 02 protected: 03 string name; 04 public: 05 virtual void introduce() { 06 cout << "I'm " << name << endl; 07 } 08 virtual void work() = 0; 09 void doDailyTask() { 10 introduce(); 11 work(); 12 } 13 }; 14 class Student: public Person { 15 private: 16 Person *pAdvisor; 17 public: 18 Student(string name, 19 Person *pAdvisor) { 20 this->name = name; 21 this->pAdvisor = pAdvisor; 22 } 23 void introduce() { 24 Person::introduce(); 25 cout << "My advisor " << endl; 26 pAdvisor->introduce(); 27 } </pre>	<pre> 28 void work() { 29 cout << "study" << endl; 30 } 31 }; 32 class Professor: public Person { 33 public: 34 Professor(string name) { 35 this->name = name; 36 } 37 void work() { 38 cout << "teach" << endl; 39 } 40 }; 41 void main() { 42 Professor p1("Hinton"); 43 Student p2("Bengio", &p1); 44 Person *p3= new Student("LeCun", 45 &p2); 46 47 p1.introduce(); 48 p1.work(); 49 50 p2.introduce(); 51 p3->introduce(); 52 53 Person *p4 = new Person(); 54 } </pre>
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- a) Are there any lines in the `main()` function that cannot be compiled? If yes, why can't they be compiled?
- b) Assume that all invalid lines of code are removed, can we change the key word `protected` in Line #2 to `private`? Explain.
- c) What is the output of this code if the invalid lines of code are removed?

Question 3. There are three types of basic electrical circuits:

	Single circuit is a circuit containing only one resistor. $R \text{ (Resistance)} = U \text{ (Voltage)} / I \text{ (Current)}$.
	Series circuit is a circuit containing more than two sub-circuits which are connected in series. $R = R_1 + R_2 + \dots + R_N$.
	Parallel circuit is a circuit containing more than two sub-circuits which are connected in parallel. $1/R = 1/R_1 + 1/R_2 + \dots + 1/R_N$.

The sub-circuit in series or parallel circuit can be either a single circuit, another series circuit, or another parallel circuit.

You are asked to do the followings by applying encapsulation, inheritance, and polymorphism:

- a) Draw a class diagram for a program to calculate circuit resistance. The design should include necessary variables and functions to:
- Construct a circuit of one type.
 - Add a sub-circuit to a Series or Parallel circuit.
 - Calculate resistance of a circuit.
- b) Write C++ code to implement the design.

*** GOOD LUCK ***